CSE 4345 – BIG DATA ANALYTICS COMPLEX ENGINEERING PROBLEM (CEP) ASSIGNMENT

EpiScope

A Big Data Surveillance, Forecasting and Response-Planning Platform for Seasonal, Tropical and Epidemic Diseases

Department of Computer Science & Engineering, Premier University, Chittagong Spring 2026

Course	CSE 4345 Big Data Analytics (3 credits)	**Instructor**	Rokan Uddin Faruqui
Assessment type	Complex Engineering Problem (group project)	**Total marks**	100 (scaled to the project weight in the course outline)
Group size	3 students	**Duration**	14 calendar days
Issued	Day 0 (announced in class)	**Final submission**	Day 14 (announced in class)
CLOs assessed	CLO2, CLO3, **CLO4**	**BAETE POs**	PO(b), (c), (d), (e), (f), (h), (i), (j), (l)

How to read this document

Section 1 is the *client brief*; that is the problem you are solving. Sections 2–4 explain why it is a Complex Engineering Problem and which trade-offs you must resolve. Section 6 lists the technical requirements, prioritised as **M** = Must, **S** = Should, **C** = Could (bonus). Sections 7–9 give the schedule, the handover package (code, report, video link) and the marking rubric. **A group that meets every M requirement well earns a solid pass. Higher grades need S items and good reasons for each design decision.**

1 Background

Bangladesh has a heavy, strongly seasonal burden of infectious disease. Dengue follows the monsoon. The 2023 season was the worst on record, with more than 300,000 hospitalised cases and over 1,700 deaths. Acute watery diarrhoea and cholera rise before the monsoon and after floods. Seasonal influenza peaks during the rainy season, and malaria is concentrated in the Chittagong Hill Tracts. On top of these recurring patterns, the country must be ready for new epidemics like COVID-19, which can overwhelm hospital capacity within weeks.

Useful data already exists, but it is spread across many systems. Health facilities report through DHIS2-style systems. Laboratories report confirmed cases. The meteorological department records rainfall and temperature. The census gives population denominators, and district stores track supplies such as IV saline, NS1 test kits and oral rehydration salts. These sources differ in format, timeliness, quality and granularity. Today most of the analysis is done after the fact in spreadsheets, so resources are often sent to a district *after* its hospitals are already full.

1.1 Client Brief (hypothetical teaching case)

From: Director, Disease Control & Preparedness Unit, Ministry of Health (MoH)
To: Data Engineering Consultancy Teams

"The Ministry needs a working prototype of a national **data platform** that turns routine health, climate and logistics data into **early warnings and an optimal response plan**. For every district, the platform must (i) detect unusual rises in dengue, diarrhoeal disease, influenza-like illness and malaria as early as possible; (ii) forecast expected cases and hospital demand for the next one to four weeks; (iii) recommend how to allocate a **limited** national stock of supplies and rapid-response teams across districts; and (iv) produce a seasonal preparedness calendar for the year ahead.

Decision-makers at the Ministry, in divisions and in districts are not data scientists. They need an interactive dashboard they can trust, that explains its recommendations, that works with incomplete and late data, and that **never exposes identifiable patient information**. The platform must run on modest infrastructure and must be able to grow to national scale."

This scenario is a teaching case written for CSE 4345. It is not an actual request from any government body, and any data you generate or

use must be labelled as synthetic or cite its public source.

1.1 Your Task

Working in a team of three, **design, build, evaluate and present** *EpiScope*: an end-to-end big data solution with the following parts:

- **T1.** a **data pipeline** that ingests heterogeneous batch and incremental sources into a layered (bronze / silver / gold) lake using Apache Spark and columnar storage;
- **T2.** **data quality, provenance and privacy** controls suitable for health data;
- **T3.** **analytics** for outbreak detection, short-term forecasting and optimal resource allocation, each evaluated quantitatively;
- **T4.** an interactive, coordinated-view **dashboard built with D3.js** that serves decision-makers at national and district level;
- **T5.** a handover package of exactly three items: the **code** (Git repository), a **technical report** (including a two-page policy brief for the Ministry), and a **video presentation** submitted as a shareable link.

2 Complex Engineering Problem Attributes

Under the BAETE accreditation manual (aligned with the Washington Accord graduate attributes), a *Complex Engineering Problem* must require in-depth engineering knowledge (P1) and must also have several of the attributes P2–P7. Table 1 shows where each attribute appears in this assignment. **Your report must include a short section explaining how your own solution dealt with each attribute.**

Table 1: Complex engineering problem attributes (P1–P7) as they appear in EpiScope.

Attribute	How it appears in this assignment
P1 Depth of knowledge (K3–K8)	Distributed storage and processing (HDFS/object stores, Spark, partitioning, skew); columnar formats; data integration and quality; time-series statistics and epidemiological surveillance methods; mathematical optimisation; visual encoding and interaction design; health-data privacy and ethics; current research literature on outbreak detection and forecasting.
P2 Conflicting requirements	Early detection (high sensitivity) vs. alert fatigue (false alarms); spatial granularity vs. patient privacy; freshness vs. compute cost; efficient allocation (send supplies where they avert most harm) vs. equity (every district gets a minimum share).
P3 Depth of analysis	No single correct solution. Detection thresholds, forecast horizons, lag structures and allocation objectives must be chosen, justified and tested against baselines using backtests and sensitivity analysis.
P4 Unfamiliar issues	Reporting delays and retrospective revisions, missing facility reports, duplicate patients across facilities, heavy data skew toward large cities, and changes in reporting practice that look like outbreaks but are not.
P5 Beyond standard codes	There is no ready-made standard for turning these combined data into a national allocation plan. You must combine guidance (WHO IDSR, HL7 FHIR/ICD-10 coding, data-protection principles) with your own engineering judgement.
P6 Stakeholders	Ministry policymakers, surveillance epidemiologists, district health officers, hospital managers, city vector-control units, the meteorological department, patients and the public, and privacy regulators all have different needs (Section 4).
P7 Interdependence	The sub-problems depend on each other. Ingestion quality affects detection; detection and forecasts feed the optimiser; the optimiser's outputs and uncertainty have to be shown clearly in the dashboard. An error in one stage carries through to the rest.

Complex engineering activities. The project also covers the activity attributes: **A1** (range of resources): people, data, compute and money under a two-week limit; **A2** (level of interaction): data, analytics and visualisation work that must fit together; **A3** (innovation): new combinations of climate, clinical and logistics signals; **A4** (consequences for society): wrong recommendations cost lives and resources; **A5** (familiarity): applying course tools in a domain outside ordinary coursework.

3 Outcome Mapping

Table 2: CLO – PO – knowledge-profile mapping and the evidence used to assess it.

CLO	Statement and level in this CEP	BAETE PO / K	Evidence
CLO2	*Analyse* the challenges of big health data: heterogeneity, quality, late data, skew, privacy (Bloom C4)	PO(b) Problem analysis; K4, K8	Proposal, stakeholder and conflict analysis, data-quality report
CLO3	*Evaluate* the performance of big data techniques and tools: partitioning, formats, joins, scaling (C5)	PO(d) Investigation; PO(e) Modern tools; K6	Scalability experiment, optimisation before/after, tool justification
CLO4	*Create* an end-to-end pipeline, analytics and D3.js dashboard that solve the brief (C6)	PO(c) Design of solutions; PO(e); K5, K6	Code, dashboard, analytics evaluation, video demonstration
CLO4	Consider societal, ethical and privacy impacts of the solution (A3, *valuing*)	PO(f) Engineer and society; PO(h) Ethics; K7	Policy brief, privacy design, ethics statement
CLO4	Work in a team and communicate to technical and non-technical audiences (A4)	PO(i) Teamwork; PO(j) Communication	Contribution matrix, individual video segments, report

4 Stakeholders and Conflicting Requirements

Stakeholder	Primary need	Typical concern
MoH policymakers	National situation at a glance; defensible allocation of a fixed budget	"Why did district X get more than Y?"
Surveillance epidemiologists	Reliable signals, method transparency, adjustable thresholds	Reporting artefacts treated as outbreaks
District health officers	Early local warning; their own district's forecast and stock	Alert fatigue; delayed supplies
Hospital managers	Bed and consumable demand forecasts	Forecasts without uncertainty bands
City vector-control units	Hotspots and timing for larval-control campaigns	Map at too coarse a resolution
Patients and public	Timely response; no misuse of personal data	Re-identification; stigma
Privacy regulator	Minimal data, purpose limitation, suppression of small counts	Line-level data leaving secure zones

Conflicts you must resolve and justify explicitly (with numbers, charts or experiments):

C1. Sensitivity vs. specificity. A lower alert threshold detects outbreaks earlier but produces more false alarms. Plot this trade-off and choose an operating point.

C2. Granularity vs. privacy. Sub-district (upazila) maps are more actionable, but small counts can identify patients. Define and enforce a suppression or aggregation policy.

C3. Efficiency vs. equity. Minimising total unmet demand can leave low-risk districts with nothing. Show an efficiency–equity curve across budget levels.

C4. Freshness vs. cost and complexity. Justify your processing cadence (daily batch, micro-batch or streaming) against the decisions it supports.

C5. Accuracy vs. interpretability. A complex model may forecast slightly better but be harder for officials to trust. Justify your choice.

5 Data Landscape

Real line-level patient data cannot be released for coursework. You will therefore **combine public open datasets with a realistic synthetic data generator that your team builds** to the specification below. Using public data for denominators, geography and climate ties the synthetic epidemiology to real seasonal and spatial patterns.

5.1 Sources

ID	Dataset	Type	Source and notes
S1	Administrative boundaries (district ADM2; upazila ADM3 optional)	Real, GeoJSON	geoBoundaries (`https://www.geoboundaries.org`) or HDX (`https://data.humdata.org`). Simplify geometries for the browser (e.g. with mapshaper/TopoJSON).
S2	Population by district, age band and sex	Real, CSV	BBS Population and Housing Census 2022 or WorldPop (`https://www.worldpop.org`). Used as incidence denominators.
S3	Daily weather per district (precipitation, temperature, humidity), 5 years	Real, API/JSON	Open-Meteo historical archive API (`https://open-meteo.com`), queried at district centroids. Respect rate limits; cache responses in bronze.
S4	Health-facility encounter line list	**Synthetic**, CSV/JSON	DHIS2-style feed generated to the specification in Section 5.2. This is the high-volume source.
S5	Facility registry	Synthetic, CSV	At least 600 facilities (hospitals, upazila health complexes, clinics) with location, type, beds and ICU beds.
S6	Daily stock and bed-occupancy snapshots	Synthetic, JSON	Per facility: IV saline, NS1 kits, ORS, antimalarials, occupied beds. Arrives as an incremental feed.
S7	Historical dengue reference (optional)	Real, CSV	OpenDengue (`https://opendengue.org`), for checking that your synthetic seasonality is realistic.

5.2 Synthetic Encounter Generator Specification (S4–S6)

The generator must be parameterised by a **scale factor** `SF` and a random seed, and must be **deterministic** for a given seed. At `SF=1` it must produce at least **25 million encounter records** for 2021–2025 across all 64 districts.

Encounter field	Description
`encounter_id`	Unique string; about 2% deliberately duplicated (resubmitted records)
`patient_id`	Synthetic national ID (bronze only; must be salted and hashed before silver)
`facility_id`	Foreign key to S5; about 0.5% orphan keys
`onset_date`, `report_date`	Reporting delay drawn from a right-skewed distribution (0–21 days); fewer reports on weekends and holidays
`age_years`, `sex`	Consistent with the S2 population structure; about 1% invalid values (e.g. age 212)
`icd10_code`	A90/A91 (dengue), A00/A09 (cholera, diarrhoea), J10/J11 (influenza/ILI), B50–B54 (malaria), other background codes
`lab_confirmed`, `admitted`, `outcome`	Severity cascade consistent with disease (e.g. dengue admission rate rises during outbreaks)

Epidemiological realism (required).

- *Seasonality:* each disease has its own annual cycle; district amplitude varies (e.g. malaria mostly in hill districts, dengue concentrated in large cities).
- *Climate drivers:* dengue incidence responds to rainfall and temperature with a lag of roughly 4–8 weeks; diarrhoeal disease spikes after extreme rainfall; drive these from the real S3 weather.
- *Overdispersion:* use negative-binomial (or Poisson–gamma) noise, not plain Poisson.
- *Injected outbreaks:* at least 30 outbreaks of different size, duration and location, including a slow-growing one and a nationwide novel-respiratory surge. Record them in a separate `ground_truth_outbreaks` file that the detection code **must never read**.

- *Artefacts:* at least one reporting artefact (e.g. a facility that goes silent for three weeks, then back-fills) that a naive detector will wrongly flag.

Hidden test drop (reproducibility check)

After submission the instructor will clone your repository, follow *only* your `README.md`, and run your pipeline on a hidden 14-day incremental drop in the *same schema*. The drop contains unannounced outbreak events and quality defects. Your code must ingest it **without modification** and regenerate the alerts, allocation plan and dashboard data. Stick strictly to the schema contract above, and do not hard-code dates, file names or row counts.

6 Technical Requirements

6.1 Reference Architecture

Figure 1 is a *reference*, not a prescription. You may substitute components if you justify the change in your report.

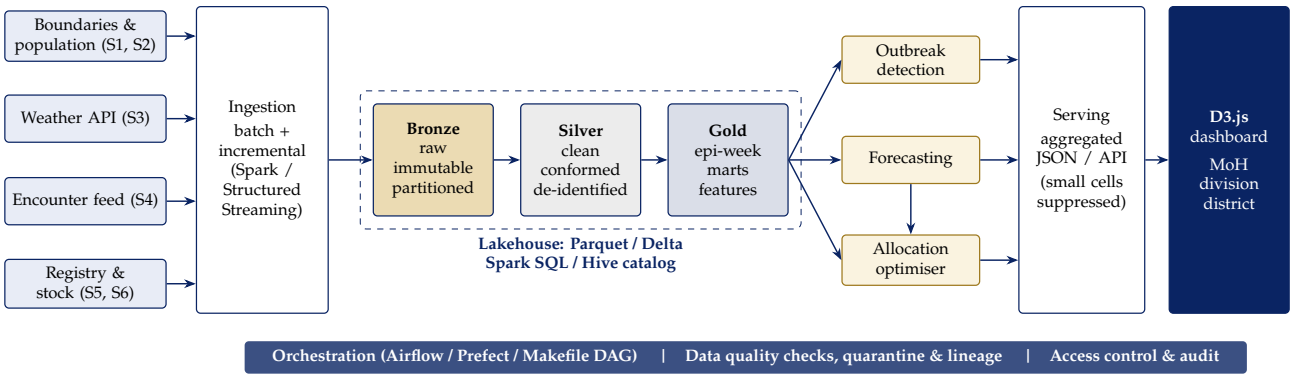


Figure 1: EpiScope reference architecture (medallion lakehouse + analytics + D3.js serving layer).

6.2 Requirement Catalogue

Requirement IDs must be traceable in your report and repository: cite the ID in commit messages, issues or README sections.

ID	Requirement	Pri.
(A) Ingestion and storage		
ING-1	Build the generator (Section 5.2) and ingest S1–S6 into a **bronze** layer without modification, adding ingestion metadata (source, load time, batch ID, file checksum).	M
ING-2	Use **Apache Spark** (PySpark or Scala) for all heavy transformations. Pandas is allowed only for small reference data or final aggregated outputs.	M
ING-3	Store silver and gold as **Parquet** (or Delta/Iceberg) **partitioned** by a justified key (e.g. epi-year/epi-week and division). Register tables in a Spark SQL or Hive catalog.	M
ING-4	**Incremental, idempotent** daily loads: re-running a day must not create duplicates, and late records (up to 21 days) must update the affected epi-weeks.	M
ING-5	Run the storage layer on HDFS (e.g. Docker Hadoop) or an S3-compatible object store (e.g. MinIO) rather than only the local file system.	S
ING-6	Near-real-time path: Spark Structured Streaming over a file source or Kafka that refreshes alerts within 5 minutes of arrival.	S
ING-7	ACID table format (Delta Lake or Apache Iceberg) with time travel, used to reproduce "what did we know on date t" for backtests.	C
(B) Data quality, provenance and privacy		
DQ-1	At least **12 automated quality rules** (schema, nulls, ranges, referential integrity, duplicates, `onset_date` $\leq$ `report_date`, valid ICD-10 and geography). Failing rows go to a **quarantine** table with a reason code.	M

ID	Requirement (cont.)	Pri.
DQ-2	A per-load data-quality report (rows in/out, rule pass rates, quarantined counts) stored as a gold table and shown on the dashboard.	M
DQ-3	Lineage: every gold record can be traced to its bronze batch IDs.	M
DQ-4	Cross-facility patient de-duplication (same hashed ID and compatible attributes within a 7-day window) with measured precision on injected duplicates.	S
PRV-1	Direct identifiers never go beyond bronze: salted hash (salt kept outside the repository), dates of birth reduced to age bands.	M
PRV-2	**Small-cell suppression**: no count from 1 to 4 appears in any dashboard output or export; totals must not allow suppressed cells to be recovered by subtraction.	M
PRV-3	Role-based views: national users see all districts; a district user sees their own district in detail and others only in aggregate.	S
PRV-4	A k-anonymity check on any record-level extract, and a written discussion of differential privacy for public releases.	C
(C) Analytics		
AN-1	**Outbreak detection** per district and disease, using at least one established method (EARS C1/C2/C3, CUSUM, or Farrington flexible), implemented at scale in Spark (window functions or grouped `applyInPandas`).	M
AN-2	Evaluate detection against `ground_truth_outbreaks`: sensitivity, false alerts per district-year and median **days to detection**. Show a threshold trade-off curve (conflict C1).	M
AN-3	Compare two detection methods, and show how your pipeline avoids alerting on the injected reporting artefact.	S
AN-4	**Forecasting**: 1–4 week-ahead weekly counts per district for at least **two** diseases, using lagged weather covariates. Train per-district models in parallel with Spark.	M
AN-5	Rolling-origin backtest over the last 52 weeks against a **seasonal-naïve baseline**: report MAE and 80% prediction-interval coverage; the weighted interval score (WIS) is recommended.	M
AN-6	Convert case forecasts to **resource demand** (beds, IV saline, test kits) with stated conversion assumptions and uncertainty.	M
AN-7	**Allocation optimiser**: formulate and solve a linear or integer programme (see Section 6.3) for a fixed national budget; compare with a population-proportional baseline.	M
AN-8	Efficiency–equity curve across at least 5 budget levels, with a recommended policy (conflict C3).	S
AN-9	**Seasonal preparedness calendar** for the next year, per division and disease group, showing when to pre-position supplies and start prevention campaigns, derived from seasonal decomposition of the data.	M
AN-10	Epidemic scenario module: a compartmental (SEIR) model calibrated on the injected novel-respiratory surge, with what-if intervention scenarios.	C
(D) Visualisation with D3.js		
VIS-1	Dashboard built with **D3.js v7** (d3-geo, d3-sankey and TopoJSON helpers allowed). Plotly, Chart.js, Vega-Lite, Tableau and Power BI must **not** be used for the required views.	M
VIS-2	At least **five coordinated views**: (1) district choropleth of incidence per 100,000 with alert markers; (2) time series with brush/zoom showing observed values, forecast and prediction bands, and alert thresholds; (3) seasonal heatmap (epi-week × year or district × epi-week); (4) allocation view comparing forecast demand, current stock and recommended allocation; (5) ranked early-warning table.	M
VIS-3	**Linked interaction**: selecting a district, disease or time range in one view filters or highlights all others. Tooltips, legends and axis labels are required.	M
VIS-4	**Scale-aware design**: the browser loads only aggregated gold outputs (total at most 10 MB), never line-level data. Justify your aggregation levels.	M
VIS-5	What-if panel: sliders for budget, detection threshold and equity floor that update allocations and alerts (from a precomputed scenario grid or a lightweight API).	S
VIS-6	Accessibility and honesty: colour-blind-safe palettes, uncertainty always shown, suppressed cells clearly marked, data-freshness timestamp on every view.	S
VIS-7	Mobile-friendly district-officer view, or animated playback of an outbreak's spatial spread.	C

ID	Requirement (cont.)	Pri.
(E) Engineering quality and performance		
ENG-1	**Reproducibility**: from a clean clone, one documented command (e.g. `make all` or `docker compose up`) rebuilds the lake, analytics and dashboard data. Pin all dependency versions.	**M**
ENG-2	**Scalability experiment**: run the full pipeline at `SF` $\in \{0.25, 0.5, 1\}$ and report runtime per stage, output sizes and hardware. At `SF=1` the pipeline must finish in 45 minutes or less on a 16 GB laptop or a free cloud notebook tier.	**M**
ENG-3	At least **two Spark optimisations** with before/after evidence (e.g. broadcast join of the registry, partition pruning, handling Dhaka-driven skew with salting or AQE, file compaction, caching). Include Spark UI screenshots.	**M**
ENG-4	Automated tests for at least the quality rules, the de-identification step and the optimiser constraints; orchestrated DAG with retries.	**S**

6.3 Optimisation Model Guidance (AN-7)

The following is a starting formulation. You may extend it with integrality, transport lead times or multiple periods, provided you justify the extension. Let D be the set of districts, R the set of resources, $\hat{y}_{d,r}$ the forecast demand for resource r in district d over the planning horizon (use an upper prediction quantile), $s_{d,r}$ the current stock, c_r the unit cost, B the budget, w_d a severity/vulnerability weight and p_d the population share. With decision variables $x_{d,r} \geq 0$ (allocation) and $u_{d,r} \geq 0$ (unmet demand):

$$\min_{x,u} \sum_{d \in D} \sum_{r \in R} w_d \, u_{d,r}$$

$$\begin{aligned}
\text{s.t.}\quad & u_{d,r} \;\geq\; \hat{y}_{d,r} - s_{d,r} - x_{d,r} && \forall d, r \quad \text{(unmet demand)} \\
& \textstyle\sum_d \sum_r c_r \, x_{d,r} \;\leq\; B && \text{(budget)} \\
& x_{d,r} \;\geq\; \alpha \, p_d \textstyle\sum_{d'} x_{d',r} && \forall d, r \quad \text{(equity floor, } 0 \leq \alpha < 1\text{)} \\
& \textstyle\sum_{d \in \mathrm{div}(k)} \sum_r v_r \, x_{d,r} \;\leq\; T_k && \forall \text{ divisions } k \quad \text{(transport capacity)}
\end{aligned}$$

Solve with an open-source solver such as PuLP/CBC, OR-Tools or SciPy HiGHS. The optimisation itself is small; the *big data* work is producing $\hat{y}$ and s reliably at scale. Explain what α means for policy (conflict C3).

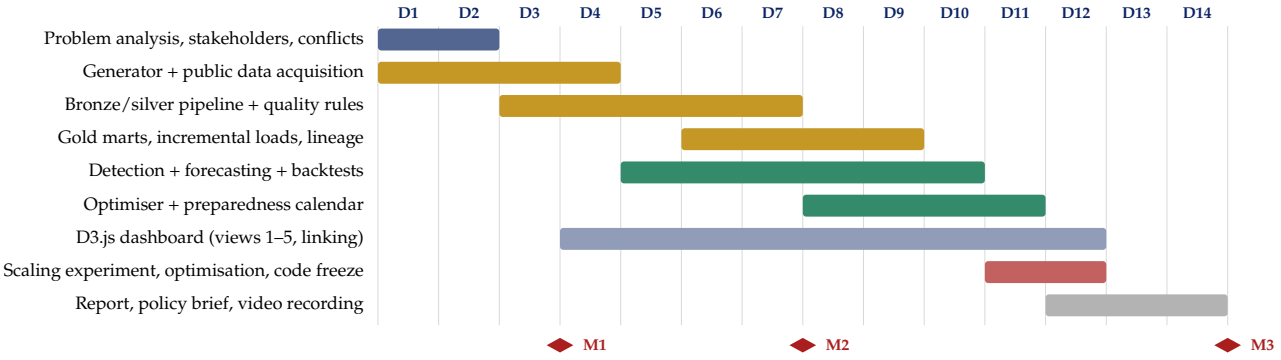
7 Team Organisation and Two-Week Plan

7.1 Roles

Each member takes *lead* responsibility for one role, but **every member must be able to explain every component** and presents their own lead area in the video. Every member must also make meaningful commits outside their own role.

Role	Lead responsibilities	Primary requirements
Data Engineer	Generator, ingestion, lakehouse layout, incremental loads, quality rules, lineage, orchestration, scalability experiment	ING-*, DQ-*, ENG-1–3
Analytics Engineer	Detection, forecasting, demand conversion, optimiser, preparedness calendar, evaluation and backtests	AN-*
Visualisation Engineer	Gold-to-serving exports, privacy suppression in outputs, D3.js dashboard, what-if panel, usability check with at least 2 non-CS users	VIS-*, PRV-2, PRV-3

7.2 Schedule and Milestones



Milestone	Due	What must be ready
M1 Proposal	End of Day 3	`docs/proposal.md` committed (at most 2 pages: problem restatement, stakeholder and conflict analysis C1–C5, planned architecture and tools with justification, chosen diseases and methods, role assignment, risk register with at least 5 risks); repository shared with the instructor, with all three members committing; generator schema committed.
M2 Checkpoint	End of Day 7	Git tag `checkpoint`: generator at `SF=0.25` flowing bronze → silver → gold; quality report; first D3 view reading gold output; updated risk log. The instructor reviews the tag and gives written feedback (formative, not graded).
M3 Handover	End of Day 14	The three handover items H1–H3 (Section 8). The repository must carry the tag `v1.0`; commits after the deadline are ignored.

8 Handover Package

Each group hands over **exactly three items** by the M3 deadline. Submit them on the course LMS as a single form containing the repository URL, the report PDF and the video link. Also put the report PDF in `docs/` and the video link at the top of `README.md`.

Item	Specification
H1 Code	Git repository (GitHub or GitLab), private and shared with the instructor, with the final state tagged `v1.0`. It must contain: • `README.md`: video link, prerequisites, **one-command run** (ENG-1), how to run the hidden test drop, architecture summary, and a requirement-ID traceability table (ID → file/notebook → report section); • `generator/`, `pipeline/`, `analytics/`, `serving/`, `dashboard/`, `tests/`, `docs/` (layout in Appendix A); • pinned environment (`requirements.txt` or `environment.yml`, `docker-compose.yml`), `AI_USAGE.md` and `docs/proposal.md` (M1); • **no** data files above 50 MB, and **no** secrets, salts or API keys (use `.env.example`); • a commit history that shows incremental contributions from all three members throughout the two weeks. The D3.js dashboard must run locally from the repository with one command (e.g. `npm run serve` or `python -m http.server`).

Item	Specification (cont.)
H2 Report	One PDF, IEEE conference format, **at most 15 pages** plus appendices. The cover page lists the group number, member names and IDs, the repository URL and the video link. Required sections: (1) Introduction and problem analysis; (2) CEP attribute analysis P1–P7 and how conflicts C1–C5 were resolved; (3) Architecture and design decisions, with alternatives considered; (4) Data generation, quality, lineage and privacy; (5) Analytics methods (detection, forecasting, demand conversion, optimisation, calendar); (6) Evaluation: detection, forecasting backtest, allocation vs. baseline, scalability and Spark optimisations; (7) Dashboard design rationale and usability findings; (8) Ethical, societal and sustainability considerations; (9) Limitations and future work; references. **Mandatory appendices** (not counted in the page limit): **A.** Two-page *policy brief* for the Director, written for a non-technical reader: current situation, top 3 recommendations with expected impact, how to read the dashboard, confidence and limitations, data-protection assurance. No code and no jargon. **B.** *Contribution matrix* (Appendix B), agreed and signed by all members. **C.** Half-page *individual reflection* from each member.
H3 Video presentation (shareable link)	• **Length**: 15–20 minutes; at least 1080p, and code, terminal and dashboard text must be readable. • **Hosting**: YouTube (*Unlisted*) or Google Drive/OneDrive set to "anyone with the link can view". The link must work **without sign-in or access requests** and stay available until results are published. An unplayable or restricted link counts as a missing submission. • **Every member presents** their lead area for at least 4 minutes, with their voice and their name on screen (camera recommended). This segment is graded individually (R8). • **Required chapters** (put timestamps in the video description or README): (1) problem, stakeholders and conflicts; (2) architecture walkthrough; (3) *recorded pipeline run*: a clean run plus one incremental day, including a re-run showing idempotency (no duplicates) and a quarantine example; (4) analytics results against baselines; (5) *dashboard walkthrough* demonstrating linked interaction across all required views and the what-if panel if implemented; (6) recommendations, ethics and limitations. • Long jobs may be sped up, but show the terminal timestamps. Do not use edits that hide failures. Results shown in the video must match the tagged `v1.0` code.

Authorship verification

Grading is based on the three handover items. If the commit history, contribution matrix or video raise doubts about who did the work, the instructor may call a short oral verification (10 minutes per member). Marks for the affected criteria may then be adjusted.

9 Assessment

9.1 Mark Distribution

#	Criterion	Marks	CLO	PO	Evidence
Group component (70)					
R1	Problem analysis: requirements, stakeholders, conflicts, CEP attribute analysis	8	CLO2	(b)	H2 §1–2
R2	Data engineering: generator, ingestion, lakehouse, incremental loads, quality, lineage, privacy controls	16	CLO3, 4	(c), (e)	H1, H2, test drop
R3	Analytics: detection, forecasting, optimisation, calendar, with rigorous evaluation	16	CLO4	(c), (d)	H1, H2 §5–6
R4	D3.js dashboard: coordinated views, interaction, scale-aware and honest design	12	CLO4	(e), (j)	H1, H3
R5	Performance investigation: scaling experiment and Spark optimisations	6	CLO3	(d), (e)	H2 §6, Spark UI
R6	Societal and ethical impact: privacy design, equity, policy brief	5	CLO4	(f), (h)	H2 §8, App. A
R7	Communication: report quality, clarity and structure of the group video	7	CLO4	(j)	H2, H3
Individual component (30)					
R8	Individual video segment: explanation of design, code and trade-offs in own lead area	15	CLO2–4	(a), (j)	H3
R9	Individual contribution: commit evidence, contribution matrix, reflection	15	CLO4	(i), (l)	H1, H2 App. B–C
	Total	**100**			
	Bonus: well-executed **C** requirements (at most +5; the total is capped at 100)	+5	CLO4	(c), (e)	H1, H3

Individual moderation. Each member's group mark G (out of 70) is multiplied by a contribution factor $\pi_i = \mathrm{clip}(3\,\sigma_i/100,\ 0.8,\ 1.1)$, where σ_i is the member's agreed percentage share in the contribution matrix (shares add up to 100, so equal shares give $\pi_i = 1$). The final mark is $\min(70, \pi_i G) + R8 + R9$. If the commit history clearly contradicts the matrix, the instructor may adjust π_i after an authorship verification.

9.2 Analytic Rubric

Criterion	Excellent (90–100%)	Proficient (70–89%)	Developing (50–69%)	Inadequate (<50%)
R1 Problem analysis	All conflicts C1–C5 analysed with stakeholder evidence; P1–P7 mapped to specific design decisions; realistic risk register.	Most conflicts analysed; P1–P7 addressed but partly generic.	Conflicts listed but not analysed; CEP mapping superficial.	Problem restated without analysis; no stakeholder view.
R2 Data engineering	All **M** and most **S** items; idempotent incremental loads proven by tests; strong quality and lineage; privacy enforced by design; hidden test drop runs from the README without changes.	All **M** items; minor gaps in late-data handling or lineage.	Pipeline works as a one-off batch only, or the test drop needs manual fixes; few quality rules; privacy partially enforced.	Spark not used meaningfully, no layering, or identifiable data leaks into outputs.

Criterion	Excellent (90–100%)	Proficient (70–89%)	Developing (50–69%)	Inadequate (<50%)
R3 Analytics	Methods justified from literature; clearly beats baselines with proper backtests; artefact handled; efficiency–equity analysis informs a clear recommendation.	Required methods implemented and evaluated against baselines; limited sensitivity analysis.	Methods implemented but evaluation is weak, in-sample only, or has no baseline.	Missing detection, forecasting or optimisation, or results not evaluated.
R4 D3.js dashboard	Five or more well-linked views; uncertainty and suppression shown clearly; fast; tested with users; tells a coherent decision story in the video walkthrough.	Five linked views with correct encodings; minor usability issues.	Views present but weakly linked, or encoding and labelling errors.	Fewer than 3 views, not built with D3.js, or loads line-level data.
R5 Performance	Clean scaling study with explanation of bottlenecks from the Spark UI; two or more optimisations with quantified gains and root-cause reasoning.	Scaling study and optimisations present; explanations partly generic.	Runtime reported for a single scale; optimisations claimed without evidence.	No performance investigation.
R6 Societal impact	Privacy, equity and misuse risks analysed concretely; policy brief is actionable and honest about uncertainty.	Relevant issues discussed; brief mostly suitable for a non-technical reader.	Generic ethics discussion; brief too technical.	Ethics or brief missing.
R7 Communication	Report is clear, well structured and evidence-based; video is well paced, follows all required chapters and shows the real system working.	Clear report and video with minor gaps (e.g. one chapter thin).	Disorganised report; video mostly slides with little of the running system.	Report or video substantially incomplete, over length, or link not accessible.
R8 Video segment	Explains design, code and trade-offs of own area precisely, with evidence, and clearly connects it to the other components.	Clear, correct explanation of own area; limited links to the rest of the system.	Mostly describes what was done, not why; some technical errors.	Segment missing, under 4 minutes, read from a script without understanding, or not by the member.
R9 Contribution	Substantial, sustained commits across roles; agreed share consistent with Git evidence; insightful reflection.	Consistent contribution to own role.	Uneven or last-minute contribution.	Minimal or unverifiable contribution.

10 Rules, Resources and Integrity

- **Compute.** Everything must run on a student laptop with 16 GB RAM using Spark in local mode or Docker, or on a free tier such as Google Colab or Databricks Free Edition. No paid cloud resources are needed or expected.
- **Permitted tools.** Any open-source library that fits the requirements. Restrictions: VIS-1 for visualisation, and Spark for heavy processing (ING-2). Cite every third-party code snippet in the repository.
- **Generative AI.** AI coding assistants are allowed. You must (i) add an `AI_USAGE.md` file listing tools, purposes and representative prompts, and (ii) be able to explain any part of the code if an authorship verification is called. Code you cannot explain earns no credit.
- **Plagiarism.** Copying between groups, or reusing public projects without attribution, will be dealt with under the university's academic integrity policy. Similarity checks will be run on reports and code.
- **Late submission.** 10% of the group mark is deducted per day, for up to 3 days. After that, the submission is not graded unless an extension was approved in advance.
- **Data ethics.** Do not scrape social media or any source containing personal data. Use only the public datasets listed (or equivalents with a clear licence) and your synthetic data.

Recommended Reading

[1] L. Hutwagner, W. Thompson, G. M. Seeman, T. Treadwell, "The bioterrorism preparedness and response Early Aberration Reporting System (EARS)," *J. Urban Health*, 80(1), 2003.

[2] C. P. Farrington, N. J. Andrews, A. D. Beale, M. A. Catchpole, "A statistical algorithm for the early detection of outbreaks of infectious disease," *J. Royal Statistical Society A*, 159(3), 1996.

[3] J. Bracher, E. L. Ray, T. Gneiting, N. G. Reich, "Evaluating epidemic forecasts in an interval format," *PLoS Computational Biology*, 17(2), 2021.

[4] D. Lazer, R. Kennedy, G. King, A. Vespignani, "The parable of Google Flu: traps in big data analysis," *Science*, 343(6176), 2014.

[5] M. Zaharia et al., "Apache Spark: A unified engine for big data processing," *Commun. ACM*, 59(11), 2016.

[6] M. Armbrust et al., "Delta Lake: High-performance ACID table storage over cloud object stores," *PVLDB*, 13(12), 2020.

[7] M. Bostock, V. Ogievetsky, J. Heer, "D^3: Data-Driven Documents," *IEEE TVCG*, 17(12), 2011.

[8] L. Sweeney, "k-anonymity: A model for protecting privacy," *Int. J. Uncertainty, Fuzziness and Knowledge-Based Systems*, 10(5), 2002.

[9] World Health Organization, *Technical Guidelines for Integrated Disease Surveillance and Response (IDSR)*, 3rd ed.

[10] Course texts: [TW] White, *Hadoop: The Definitive Guide*; [LRU] Leskovec, Rajaraman & Ullman, *Mining of Massive Datasets*; [TE] Erl et al., *Big Data Fundamentals*.

A Suggested Repository Layout

```
episcope/
  README.md                                          setup, one-command run, requirement traceability
  AI_USAGE.md   docker-compose.yml   Makefile   requirements.txt
  generator/                                         seeded synthetic S4–S6, ground-truth outbreaks
  pipeline/bronze/   pipeline/silver/   pipeline/gold/   pipeline/quality/
  analytics/detection/   analytics/forecast/   analytics/optimise/   analytics/calendar/
  serving/                                           gold → suppressed JSON exports or small API
  dashboard/                                         index.html, D3 modules, TopoJSON
  orchestration/   tests/   docs/ (report, brief, slides, experiment logs)
```

B Contribution Matrix (Report Appendix B)

Copy this table into Appendix B of the report. Enter the *lead* (L), *support* (S) or no involvement (–) for each member in each area, then agree on an overall percentage share. All members sign to confirm that they agree.

Work area	Member 1	Member 2	Member 3
Problem analysis, proposal, conflict analysis			
Synthetic generator and public-data acquisition			
Ingestion, lakehouse layers, incremental loads			
Data quality, lineage and privacy controls			
Outbreak detection and evaluation			
Forecasting, backtesting and demand conversion			
Optimisation model and preparedness calendar			
Serving layer and D3.js dashboard			
Scalability experiment and Spark optimisations			
Report and policy brief			
Video production			
Agreed overall share σ_i (%, total 100)			
Commits / lines changed (from `git shortlog -sn`)			
Signature			